

UNIT - II

2

Data Storage and Cloud Computing

Syllabus

Data Storage : Introduction to Enterprise Data Storage, Direct Attached Storage, Storage Area Network, Network Attached Storage, Data Storage Management, File System, Cloud Data Stores, Using Grids for Data Storage. **Cloud Storage :** Data Management, Provisioning Cloud storage, Data Intensive Technologies for Cloud Computing. **Cloud Storage from LANs to WANs :** Cloud Characteristics, Distributed Data Storage

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- 2.1 Introduction to Enterprise Data Storage
- 2.2 Data Storage Management
- 2.3 File System **March-19,20,** Marks 6
- 2.4 Cloud Data Stores **Dec.-16,17, March-19,** Marks 5
- 2.5 Using Grids for Data Storage
- 2.6 Cloud Storage
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- 2.8 Multiple Choice Questions

2.1 Introduction to Enterprise Data Storage

- Enterprise storage is a centralized repository for business - critical information that provides data sharing, data management and data protection across multiple computer systems.
- Enterprise storage may include a Storage Area Network (SAN), Network - Attached Storage (NAS) devices and Direct Attached Storage (DAS).

2.1.1 Direct Attached Storage

- Direct - Attached Storage (DAS) is hard disk drives or solid - state drives connected directly inside or outside (in a storage enclosure) to a single computer or server that cannot be accessed by other computers or servers. DAS is not networked through Ethernet or FC switches.
- The storage device may include one or more drives built into a server and with an appropriate host bus adapter, may be configured as a RAID array.
- Examples of DAS include hard drives, optical disc drives and storage on external drives.
- One advantage of DAS storage is its low initial cost. A key disadvantage of DAS storage is its limited scalability. A Host Bus Adaptor can only support a limited number of drives.
- Direct attached storage tends to be less expensive than NAS and SAN implementations, but lacks scalability and flexibility.
- DAS are of two types : Internal DAS and external DAS.
 1. **Internal DAS** : Storage device is internally connected to the host. The connection between host and storage device is serial or parallel bus.
 2. **External DAS** : Server connects directly to the external storage device.

Advantages of DAS :

1. DAS is ideal for localized file sharing in environments with a single server or a few servers.
2. DAS devices can offer block-level access or file-level access.
3. DAS also offers ease of management and administration.
4. DAS can still be used locally to store less critical data.
5. Initial cost of DAS is lower than NAS.

Disadvantages of DAS :

1. DAS is limited in its scalability.
2. DAS is limited to dedicated servers.

3. Unused resources cannot be reallocated easily.
4. If the server the device is attached to is down for any reason, the data stored on attached DAS devices is inaccessible.

2.1.2 Storage Area Network

- Storage Area Network (SAN) is a dedicated high-performance network or subnetwork dedicated to storage that is independent of an organization's common user network.
- It interconnects pools of disk or solid - state storage and shares it to multiple servers so each one can access data as if it was directly attached. A storage area network is a dedicated network that provides access to consolidated, block level data storage.
- SANs are primarily used to enhance storage devices, such as disk arrays, tape libraries and accessible to servers so that the devices appear like locally attached devices to the operating system.
- SAN systems provide a simple, untyped, fixed-size (block), memory-like interface for manipulating non-volatile magnetic media. SAN is used when multiple servers need access to the same data (Example : Databases).
- A SAN is a high - speed network of storage devices, separate from the local area network that also connects those storage devices with servers. It provides block - level storage that can be accessed by the applications running on any and all servers connected to the SAN.
- SAN storage devices can include tape libraries and more commonly, disk-based devices, like RAID hardware.
- Fig. 2.1.1 shows storage area network.

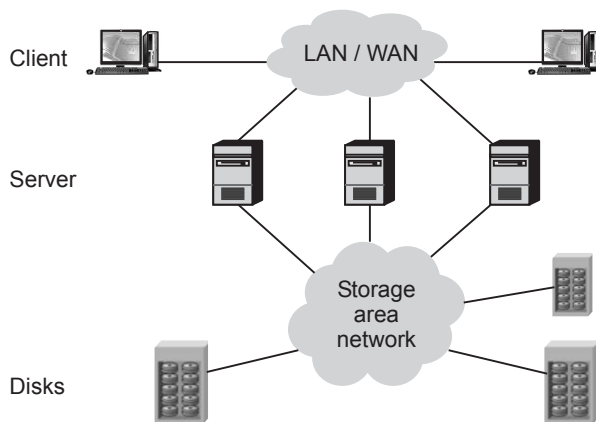


Fig. 2.1.1 Storage area network

- SANs are primarily used to enhance storage devices, such as disk arrays, tape libraries and accessible to servers so that the devices appear like locally attached devices to the operating system
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- A storage area network is designed for managing very large amounts of network storage. For large organizations with many servers that need access to the same data, SAN offers better performance and flexibility than DAS or NAS.
- Storage area network is defined as a set of interconnected devices and servers that are connected to a common communication and data transfer infrastructure such as fiber channel. The purpose of the SAN is to allow multiple servers access to a pool of storage in which any server can potentially access any storage unit.
- A SAN is a network designed to transfer data from servers to targets and it is alternative to directly attached target architecture or to a DAS architecture, where the storage is connected to the servers on general purpose networks.
- A SAN can be considered as an extended and shared storage bus within a data center, consisting of various storage devices and specific interfaces (e.g., fibre channel) rather than the Ethernet. SANs provide high - bandwidth block storage access over long distance via extended fiber channel links.

SAN features :

1. Network resource used exclusively for the storage.
2. SCSI protocol for end to end communication : Minimum impact for DAS to SAN migration.
3. Access to data through logical blocks and not to file.
4. Compatibility with an high number of nodes.
5. High reliability and ability to react to failures.
6. Performance allows concurrent access of disk or tape arrays by two or more servers at high speeds.
7. Availability have disaster tolerance built in, because data can be mirrored using a SAN up to 10 km or 6.2 miles away.

- Scalability like a LAN/WAN, it can use a variety of technologies. This allows easy relocation of backup data, operations, file migration and data replication between systems.

Advantages of SAN :

1. SAN support large and heterogeneous block data transfer
2. It gives data transfer reliability
3. Reduces LAN traffic
4. Configuration flexibility
5. High performance
6. High scalability
7. Centralized management
8. Multiple vendor offerings
9. Resilience to failure.

2.1.3 Network - attached Storage

- NAS is a file level computer data storage server connected to a computer network providing data access to a heterogeneous group of clients. It is specialized for serving files either by its hardware, software or configuration.
- NAS devices are storage arrays or gateways that support file - based storage protocols such as NFS and CIFS and are typically connected via an IP network.
- Components of NAS is NIC, OS, 1. CPU and memory, NFS and CIFS, Storage protocol
- Fig. 2.1.2 shows component of NAS.
(See Fig. 2.1.2 on next page)
- NAS uses industry standard storage protocols (ATA, SCSI and FC) for connecting and managing physical disk resources. Network file system and common internet file system protocols used for file sharing. Network interface card is used for providing network connectivity.
- NAS devices are the most appropriate for high - demand engineering applications like computer - aided design and software development.
- If you have both UNIX and windows users on your network and you want both groups to be able to share files, NAS devices are most suitable. NAS devices can make use of existing directories of user accounts from Windows, Netware or UNIX server.

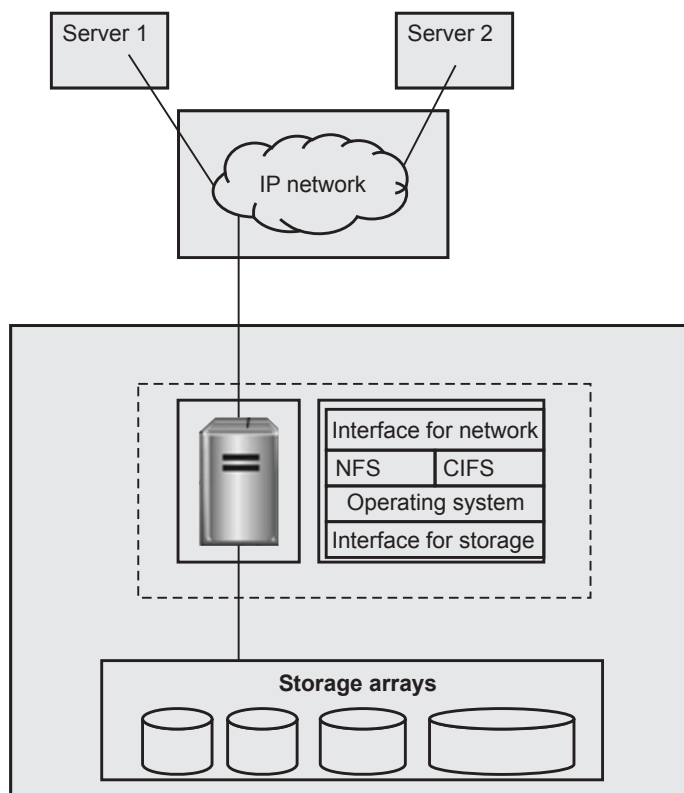


Fig. 2.1.2 NAS components

NAS File I/O :

- In NAS, there are two file level protocols which allows the file sharing in a NAS.
 1. **CIFS** : Common internet file System, this protocol allows windows clients/users to share, store and retrieve files on a NAS device
 2. **NFS** : Network file system, this protocol allows Unix clients/users to share, store and retrieve files on a NAS device
- NAS uses file level access for all types of I/O operations. File level I/O is a high level type of request that specified only file to access. In this file level storage, the storage disk is configured with a protocol such as NFS or CIFS and the files are stored and accessed from it in bulk.
- It stores files and folders and the visibility is the same to the clients accessing and to the system which stores it.
- File I/O request specifies the file and offset into the file.
 1. File system and remote file sharing : File system is a structured way of storing and organizing data files.

2. Accessing file system : File system must be mounted before it can be used. During boot process, operating system mounts the local file system. To mount any file system, two pieces of information must be specified :
 - a. A means of uniquely identifying the desired disk drive and partition, such as device file name, file system label.
 - b. A directory under which the mounted file system is to be made available
- Mount utility allows a super user to splice the root directory of a file system into the existing directory hierarchy. File system may be detached from the main hierarchy using the un-mount utility.
- NAS places a network between client and file system. NAS systems contain the file system that interprets metadata, so they do not send as much metadata across the network. NAS systems have traditionally trusted less, mainly the operating systems of their clients but not the users of these clients.
- A NAS device does not need to be located within the server but can exist anywhere in a LAN and can be made up of multiple networked NAS devices.
- NAS systems usually contain one or more hard disks, often arranged into logical, redundant storage containers or RAID arrays.
- NAS does not provide any of the activities that a server in a server - centric system typically provides, such as e-mail, authentication or file management. NAS allows more hard disk storage space to be added to a network that already utilizes servers without shutting them down for maintenance and upgrades.
- NAS systems contain one or more hard disks, often arranged into logical, redundant storage containers or RAID arrays.

NAS Implementation :

- NAS is implemented in two ways : Integrated and gateway.

1. Integrated NAS :

- Integrated NAS contains all the components of NAS so it makes the integrated NAS a self contained environment. Fig. 2.1.3 shows integrated NAS. Integrated NAS is useful when customer is looking at an integrated NAS which has its own storage.

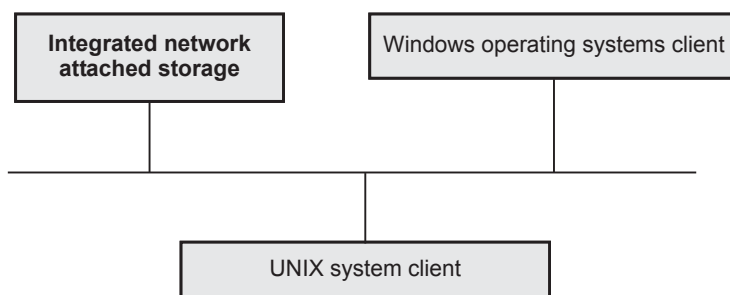


Fig. 2.1.3 Integrated NAS

- An integrated NAS solution is available with low - end device to high end solution. They are connected to the local area network and communicate with file systems through Trans - Mission Control Protocol / Internet Protocol (TCP / IP).

2. Gateway NAS :

- It is NAS device with the internal processor but without integrated storage. Instead, the NAS device connects to storage by direct attachment or by a SAN.
- In a NAS gateway, the proprietary RAID controller is replaced with an open, ANSI-standard SAN interface, usually a fibre channel host bus adapter. This allows the NAS gateway to use shared, nonproprietary storage, which provides the flexibility for an effective and scalable storage strategy.
- When deploying NAS gateways, it is important to select the right shared storage platform to house your organization's data.
- Fig. 2.1.4 shows NSA gateway.

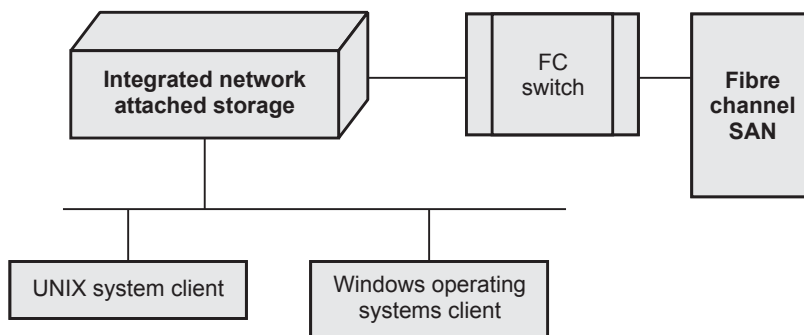


Fig. 2.1.4 Gateway NAS

- A front end to disk storage that is highly scalable. A NAS gateway functions like a diskless file server with unlimited storage. Unlike the traditional NAS device, which contains the file sharing protocol for access, the file system for storage and the disks themselves, NAS gateway contains only the file sharing protocol and the file system.
- The front end of the NAS gateway connects to the LAN over Ethernet. The back end connects to a RAID array via fibre Channel or to a fibre Channel switch, in which case the NAS gateway becomes a front end to a SAN.

Difference between Integrated NAS and Gateway NAS :

Sr. No.	Integrated NAS	Gateway NAS
1.	Integrated NAS uses fibre channel disks only.	Gateway NAS uses both fibre channel and SATA disks.
2.	Integrated NAS manages storage devices through NAS software.	Gateway NAS relies on storage array management software to manage storage devices.
3.	It uses specialized operating systems.	It uses generic operating systems.
4.	Integrated NAS allows both file and block level access to the clients.	Gateway NAS allows only block level access to the clients.

- **Benefits of NAS :**

1. NAS support NFS and CIFS file system.
2. Entry level NAS systems are quite inexpensive.
3. NAS ensure security and user authentication.
4. It helps for easy and fast communication.
5. NAS appliances already include integrated mechanisms for backup, data synchronization and data replication.
6. Simple to administrate, even remotely, i.e. from another location.
7. NAS device may support one or more RAID levels to make sure that individual disk failures do not result in loss of data.
8. NAS is a much simpler and less expensive technology.

2.1.4 Comparison between NAS and SAN

Sr. No.	NAS	SAN
1.	Machine connected with LAN may utilize NFS, CIFS or HTTP protocol to connect to a NAS	Server class devices that are equipped with SCSI and fibre channel adapters connect to a SAN.
2.	File system is managed by the NAS head unit	The SAN servers manage the file system
3.	Backups and mirrors are generated on files, not blocks	Backups and mirrors require a block by block copy operation.
4.	A NAS identifies the data by file name and byte offset, transfers file data or metadata	SAN addresses the data by logical block numbers and transfers the data in disk blocks.
5.	NAS uses TCP/IP networks	SAN uses fibre channel

2.1.5 Comparison between DAS, NAS and SAN

Sr. No.	Parameter	DAS	NAS	SAN
1.	I/O Protocol	SCSI	NFS and CIFS	SCSI
2.	Data sharing	Not supported	Supported	SAN support data sharing with the help of specialized software
3.	Capacity sharing	Manual	Yes	Yes
4.	Media	FC, parallel SCSI	Ethernet	Fiber channel
5.	Storage	Storage is directly attached by a cable to the processor	NAS device is attached to a TCP/IP-based network	Storage resides on a dedicated network

2.2 Data Storage Management

- Large number of databases is generated in private organization and government organization. It is growing out of control. Huge databases can be difficult to query, challenging to secure and maintain and cumbersome to replicate for testing and development purposes.

2.2.1 Data Storage Management Tools

- Storage monitoring software helps user to keep track of storage capacity and assigning storage to various applications accordingly. It saves time and user efficiency. Regular monitoring of the physical health, failures, and outages can help manage storage devices.
- Configuration, migration, provisioning, storage monitoring and reporting are the management level tasks. Storage Resource Management (SRM) tools include configuration tools, provisioning tools and measurement tools.
- Storage management tools include capacity management, monitoring, cloud storage management and other devices and applications to manage SAN and NAS systems. Storage management also includes backup and secondary storage tools such as snapshots and copy data management.
 1. **Configuration tools** : It handles the set-up of storage resources. These tools help to organize and manage RAID devices.
 2. **Provisioning tools** : It defines and control access to storage resources for preventing a network user from being able to use any other user's storage.
 3. **Measurement tools** : Analyse performance based on behavioural information about a storage device.

2.2.2 Storage Management Process

- Storage management covers three area : Change management, performance and capacity planning, tiering.
- Change management : Process used to request, schedule, implement and evaluate adjustments to the storage infrastructure is called change management.
- Performance and capacity planning are used to measure the performance of a system in terms of storage and utilization.
- Storage management process is as follows :
 - a. Start the process : Data is stored according to its importance and access frequency.
 - b. Data backup, restore and recovery operations : Data backups are made on a periodic basis, also restore and recovery planes are made to guide restore and recovery operations to bring backups online.
 - c. Storage resource management : Involves managing and maintaining those resources on which the backed up data us stored like storage media.

2.2.3 Cloud Provisioning

- Cloud provisioning is the allocation of a cloud provider's resources and services to a customer. It is a key feature of the cloud computing model, relating to how a customer procures cloud services and resources from a cloud provider. The growing catalog of cloud services that customers can provision includes infrastructure as a service, software as a service and platform as a service, in public or private cloud environments.
- Provisioning is the process by which a resource is prepared for use, reserved, accessed, used and then released when the transaction is completed.
- It is a key feature of the cloud computing model, relating to how a customer procures cloud services and resources from a cloud provider.
- The growing catalog of cloud services that customers can provision includes infrastructure as a service, software as a service and platform as a service, in public or private cloud environments.

Benefits of provisioning :

1. Continuous improvement of activity which relay on process measurement.
2. There is isolation of install, configure, build and customize task.
3. Ability to measure progress of all work
4. Assembly line approach to provisioning.

Characteristics of provisioning :

1. Missing and incorrect information is resolved.
 2. Product being deliver to customer before provisioning.
- The characteristics of a provision are that it is a liability where there is uncertainty as to either the timing of settlement or the amount to be settled.
 - When measuring a provision, the amount to be recognized should be the best estimate of the consideration required to settle the present obligation at the end of the reporting period.
 - The fact that it is difficult to measure a provision and that estimates have to be used does not mean that the provision is not reliably measurable.
 - **Types of cloud provisioning are as follows :**
 1. Dynamic provisioning : Cloud resources are deployed flexibly to match a customer's fluctuating demands. The deployments typically scale up to accommodate spikes in usage and scale down when demands decrease. The customer is billed on a pay-per-use basis.
 2. Self - provisioning : With user self-provisioning, also called cloud self-service, the customer buys resources from the cloud provider through a web interface or portal. This usually involves creating a user account and paying for resources with a credit card.
 3. Advance provisioning : Customer undertake contract with the provider for the required services.

2.3 File System**SPPU : March-19,20**

- File system is an interface between secondary storage device like hard disk and user application. Hard disk is a block oriented device.
- The purpose of file systems is to maintain a consistent view of storage so that we can effectively manage it. This is done in a way that allows the users to create files and directories as well as delete, open, close, read, write and/or extend the files on the device(s).
- File systems also maintain security over the files by using access control lists for a file.
- File system is a structured data representation and a set of **metadata** that describe the stored data. File system also support **disk parition**.

2.3.1 FAT File System

- File Allocation Table (FAT) file system is a simple file system originally designed for small disks and simple folder structures.
- Usually the file system operates blocks, not sector. **File system blocks** are groups of sectors that optimize storage addressing. Modern file systems generally use block sizes from 1 up to 128 sectors (512 - 65536) bytes.
- Windows Operating System use two major file systems : **FAT and NTFS**. Recently released **ReFS** file system was developed by Microsoft as a new generation file system for windows 8 servers.
- The number in FAT12, FAT16 and FAT32 stands for the number of bits used in file system block.
- Linux uses Ext2, Ext3, Ext4 file system. Ext3 file system is just an extension to Ext2 that uses transactional file write operations with journal.

2.3.2 NTFS

- NT File System (NTFS), which is also sometimes called the new technology file system, is a process that the Windows NT operating system uses for storing, organizing and finding files on a hard disk efficiently.
- Formatting a volume with the NTFS file system results in the creation of several system files and the Master File Table (MFT), which contains information about all the files and folders on the NTFS volume.
- NTFS file system supports larger file sizes and hard drives and is more secure than FAT. In NTFS, the size of cluster ranges from 512 bytes to 64 kilobytes.
- NTFS file system can help you to compress files and folders or the whole NTFS partition to save disk space so you can make best use of it.

2.3.3 Cloud File System

- Cloud storage is the abstraction, pooling and sharing of storage resources through the internet. File storage is the dominant technology used on NAS systems and is responsible for organizing data and representing it to users. Its hierarchical structure allows us to navigate data from top to bottom easily, but increases processing time.
- A file system in the cloud is a hierarchical storage system that provides shared access to file data. Users can create, delete, modify, read and write files and can organize them logically in directory trees for intuitive access.

- Cloud file storage is most appropriate for unstructured data or semi - structured data, such as documents, spreadsheets, presentations and other file - based data.
- Cloud file storage is a storage service that is delivered over the Internet, billed on a pay-per-use basis and has an architecture based on common file level protocols such as Server Message Block (SMB), Common Internet File System (CIFS) and Network File System (NFS).
- Cloud File system should be scalable enough to adopt large organizations file systems under different workloads with good performance requirements.
- Cloud file storage is a method for storing data in the cloud that provides servers and applications access to data through shared file systems. This compatibility makes cloud file storage ideal for workloads that rely on shared file systems and provides simple integration without code changes.
- A file system in the cloud is a hierarchical storage system that provides shared access to file data. Users can create, delete, modify, read, and write files and can organize them logically in directory trees for intuitive access.

Benefits of Cloud File Storage :

- Storing file data in the cloud delivers advantages in three key areas :
 1. **Scalability** : Although not every cloud file storage solution can scale, leveraging all the capabilities of the cloud, the most advanced solutions provide the ability to start with the capacity and performance you need today and grow your capacity as needed. No more over provisioning to try and anticipate future needs.
 2. **Interoperability** : Many existing applications require integration with shared file services that follow existing file system semantics. Cloud file storage solutions offer a distinct advantage as there is no new code to write to have secure, shared file access.
 3. **Budget and Resources** : Operating file services on-premises requires budget for hardware, ongoing maintenance, power, cooling, and physical space. Cloud file storage enables organizations to redeploy technical resources to other projects that bring more value to the business.

1. Ghost File System :

- Amazon Web Services (AWS) uses Ghost cloud File System (GFS). GFS run over Amazon's EC2, simpleDB web service and S3.
- GFS is elastic and cost efficient. It is highly secure. It can be mounted on server, client or access files via web page.

2. Hadoop File System :

- Hadoop Distributed File System (HDFS) is a distributed file system inspired by GFS that organizes files and stores their data on a distributed computing system.
- Hadoop is an open source software framework that supports data-intensive distributed applications, licensed under the Apache v2 license. It provides software framework for distributed processing of large datasets in real-time applications.
- Hadoop Distributed File System is a block-structured file system where each file is divided into blocks of a pre-determined size. These blocks are stored across a cluster of one or several machines.
- Blocks are the smallest continuous location on your hard drive where data is stored. Similarly, HDFS stores each file as blocks which are scattered throughout the Apache Hadoop cluster. The default size of each block is 128 MB in Apache Hadoop 2.x (64 MB in Apache Hadoop 1.x) which you can configure as per your requirement.

3. Kosmos File System (KFS) :

- KFS is an open source project written in C++ by search startup Kosmix.
- The Kosmos filesystem consists of three components :
 - a) One or multiple chunk servers that store the data on their own hard disks,
 - b) A metasever that keeps an eye on the chunk servers, and
 - c) An application that quickly gets rid of a single large file
- KFS first splits a file into handy 64 MB blocks. The filesystem distributes these chunks evenly over all attached servers, aptly referred to as block or chunk servers. The servers store the blocks on normal filesystems that belong to the host operating systems.

Review Questions

1. Draw and explain Cloud File Systems GFS/HDFS.

SPPU : March-19, In Sem., Marks 6

2. Write a note on cloud file system with architectures.

SPPU : March-20, In Sem., Marks 5

2.4 Cloud Data Stores

SPPU : Dec.-16,17, March-19

- Data store is a connection to a store of data, whether the data is stored in a database or in one or more files. The data store may be used as the source of data for a process.
- A datastore is a repository for storing, managing and distributing data sets on an enterprise level.

- Datastore data may be structured, unstructured or in another electronic format. Depending on the organization, a datastore may be classified as an application - specific datastore, operational datastore or centralized datastore.

2.4.1 Distributed Data Store

- A computer network where information is stored on more than one node, by means of data replication it is termed as Distributed Data Store. It is used to refer to a distributed database where users store information on a number of nodes or a computer network in which users store information on a number of peer network nodes.
- Distributed data system is one in which files, scripts and images are stored in more than one server or volumes rather than a single server as in traditional systems.
- Example : Google's Big Table, Amazon's Dynamo, windows Azure storage
- These types of data store are non - relational databases that searches data quickly over a large multiple node.

2.4.2 Data Store Types

2.4.2.1 BigTable

- Bigtable is a distributed storage system that is used for managing and storing structured data at Google.
- Bigtable is designed to reliably scale to petabytes of data and thousands of machine. Bigtable has multiple goals like applicability, high availability, scalability, high performance.
- It is used by approximate sixty Google project or product like Google Analytic, Google Finance, Personalized search, Writely and Google Earth.
- Bigtable is built on Google file system for storing the data for scheduling large scale data processing. It stored data in form of rows, columns and timestamp that means it maps with arbitrary string value like row key and colmn key as well as timestamp.

Features :

1. Bigtable is a distributed storage system for managing strcutured data.
2. Bigtable uses the distributed Google File System (GFS) to store log and data files.
3. A Bigtable cluster stores a number of tables. Each table consists of a set of tablets and each tablet contains all data associated with a row range.

4. Bigtable use a highly available and persistent distributed lock services called Chubby.
 5. Bigtable supports single - row transactions.
- The map is indexed by a row key, column key and a timestamp ; each value in the map is an un - interpreted array of bytes.

1. Row key :

- The row keys in a table are arbitrary strings. Every read or write of data under a single row key is atomic.
- Bigtable maintains data in lexicographic order by row key. The row range for a table is dynamically partitioned.
- Each row range is called a tablet, which is the unit of distribution and load balancing.

2. Column key :

- Column keys are grouped into sets called column families, which form the basic unit of access control. All data stored in a column family is usually of the same type.
- A column family must be created before data can be stored under any column key in that family; after a family has been created, any column key within the family can be used.

2.4.2.2 Dynamo

- Dynamo is propriety key value structured storage system. It can act as database and also distributed hash table.
- Dynamo dynamically partitions a set of keys over a set of storage nodes
- It is most powerful relational database available in WWW. Relational databases have been used a lot in retail sites, to make visitors browse and search for product easily.
- Dynamo does not support replication.
- Dynamo is used to manage the state of services that have very high reliability requirements and need tight control over the tradeoffs between availability, consistency, cost-effectiveness and performance.
- There are many services on Amazon's platform that only need primary-key access to a data store. For many services, such as those that provide best seller lists, shopping carts, customer preferences, session management, sales rank, and product catalog, the common pattern of using a relational database would lead to inefficiencies and limit scale and availability. Dynamo provides a simple primary-key only interface to meet the requirements of these applications.

- Dynamo is a completely decentralized system with minimal need for manual administration. Storage nodes can be added and removed from Dynamo without requiring any manual partitioning or redistribution.
- Compared to Bigtable, Dynamo targets applications that require only key/value access with primary focus on high availability where updates are not rejected even in the wake of network partitions or server failures.
- Dynamo stores objects associated with a key through a simple interface; it exposes two operations : get() and put().
- Dynamo treats both the key and the object supplied by the caller as an opaque array of bytes. It applies a MD5 hash on the key to generate a 128-bit identifier, which is used to determine the storage nodes that are responsible for serving the key.
- Dynamo's partitioning scheme relies on consistent hashing to distribute the load across multiple storage hosts. In consistent hashing , the output range of a hash function is treated as a fixed circular space or "ring".
- Dynamo provides eventual consistency, which allows for updates to be propagated to all replicas asynchronously.
- Dynamo uses vector clocks in order to capture causality between different versions of the same object. A vector clock is effectively a list of (node, counter) pairs. One vector clock is associated with every version of every object.
- In Dynamo, when a client wishes to update an object, it must specify which version it is updating. This is done by passing the context it obtained from an earlier read operation, which contains the vector clock information.
- In Dynamo, each storage node has three main software components: request coordination, membership and failure detection, and a local persistence engine. All these components are implemented in Java.

Review Questions

1. Write a short note on Dynamo.

SPPU : Dec.-16, 17, Marks 5

6. Draw and explain architecture of Amazon Dynamo.

SPPU : March-19, In Sem., Marks 5

2.5 Using Grids for Data Storage

- Grid computing is a distributed computing system where a group of computers are connected to create and work as one large virtual computing power, storage, database, application and service.

- Grid storage is a specific type of data storage that uses a series of interconnected nodes to accommodate versatile and scalable storage systems.
- Grid computing is based on physically distributed computer resources used cooperatively to run one or more applications. The resources may be owned by several organizations and the main goal is the efficient processing of specific applications that need access to specific resources, for example, there may be data resources held at various locations that must all be accessed by a particular application.
- **Grid-Oriented Storage (GOS)** is a dedicated data storage architecture which can be connected directly to a computational grid to support advanced data bank services and reservoirs for data that can be shared among multiple computers and end users on the grid.
- GOS is a successor of well - used Network-Attached Storage (NAS) products in the Grid Computing.
- GOS is designed to deal with long - distance, cross - domain and single - image file operations, which is typical in Grid environments. GOS behaves like a file server via the file - based GOS - FS protocol to any entity on the grid. Inspired by the success of GridFTP, GOS - FS integrates a parallel stream engine and Grid Security Infrastructure (GSI).

2.6 Cloud Storage

- Cloud storage solve problems of traditional storage. Cloud storage is a cloud computing model that stores data on the internet through a cloud computing provider who manages and operates data storage as a service.
- For solving problem of traditional storage, following practices are used :
 - a) Huge and unpredictable growth
 - b) Cost and complexity of conventional storage
 - c) Security.
- Because cloud storage is handled on a pay - as - you - go basis, the platform is built to handle rapid scaling as needed. In a local hardware scenario, additional storage and hardware must be purchased, integrated and configured by the IT.
- Cloud file storage providers have designed their infrastructure to handle client needs of all shapes and sizes, meaning that they can adapt quickly in the case of a client with an exploding user base.
- Attributes of cloud storage :
 1. **Multitenancy** : Multitenancy is when several different cloud customers are accessing the same computing resources, such as when several different companies are storing data on the same physical server.

2. **Resource pooling** : Resource pooling allows cloud providers to pool large-scale IT resources to serve multiple cloud consumers.
3. **Scalable and elastic** : Cloud storage services provide elasticity, which means you can scale capacity as your data volumes increase or dial down capacity if necessary.
4. On-demand self service
5. **Pricing based usage** : Cloud storage helps companies cut costs by eliminating in-house storage infrastructure.

2.6.1 Data Management for Cloud Storage

- Data management encompasses acquiring, storing, protecting and processing data across an organization or business unit. It helps ensure that data is validated and fully accessible to stakeholders when needed.
- Data management in the cloud is part of a broader trend of data modernization. In the past, cloud and data strategies were often managed separately. Today, they should be part of a broader unified strategy that takes advantage of the cloud's inherent elasticity, scalability and economies of scale to enable faster, more powerful business insights and new capabilities.
- The SNIA Cloud Storage Technologies Initiative (CSTI) is committed to the adoption, growth and standardization of storage in cloud infrastructures. This encompasses data services, orchestration and management, as well as the promotion of portability of data in multi - cloud environments.
- Fig. 2.6.1 shows cloud storage usage of SIRDMM model.

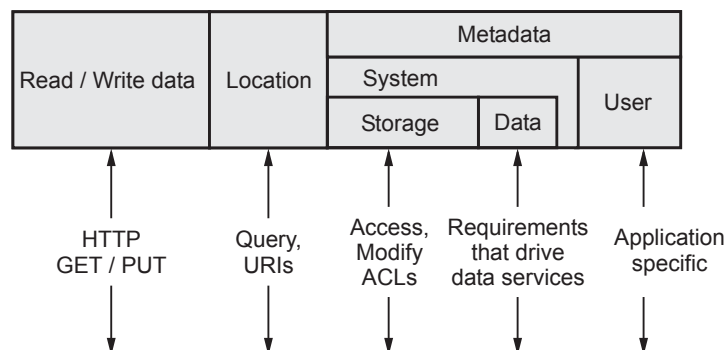


Fig. 2.6.1 Cloud storage usage of SIRDMM model

- SIRDMM model uses three types of metadata : Storage system metadata, data system metadata and user metadata.
- Storage system metadata contain information about size, access time, owner and ACL. Data system metadata contain information about encryption and retention etc.

2.6.2 Cloud Data Management Interface

- Cloud Data Management Interface is the interface for data management in cloud computing. This functional interface can be used to retrieve, create, delete and update data elements from any cloud.
- It permits the client to find out the available capabilities in cloud storage. It further helps in the management of the data and the containers place in them. It also allows the accusation between the metadata and the containers.
- CDMI features and benefits
 - a) CDMI adds powerful data management capabilities that include administration and management of user accounts, security access, monitoring and billing information.
 - b) CDMI clients can discover the capabilities of the cloud storage offering and use this information to tailor the management of containers and the data placed in them.
 - c) CDMI is an open international (ISO) standard designed to work alongside OpenStack Swift and Amazon S3 Models.
- The operations of the Cloud Data Management Interface are divided into two types by international standards :
 - a) Those that use the CDMI content type in the HTTP body
 - b) Those who do not make the use of it
- The CDMI can also be used by the management and administrative applications, it defines the means to store and retrieve the data as well as manage the data. The means by which data management is done is known as control path and the means by which data retrieval is achieved is known as the data path.
- CDMI can also manage the cloud storage properties and its capabilities are discovered by the capabilities allowed to a client.

2.6.3 Cloud Storage Requirement

1. Multi - tenancy

- A multi - tenant cloud is a cloud computing architecture that allows customers to share computing resources in a public or private cloud. Each tenant's data is isolated and remains invisible to other tenants.
- It allows multiple users to work in a software environment at the same time, each with their own separate user interface, resources and services. The multitenant application design was created to enable multiple users (tenants) to access the same application logic simultaneously.

2. Security :

- Security policy establishes a set of security rules and regulations. Often, security policies will further define how these rules and regulations are implemented and enforced.
- Securing data can be done using encryption, authentication and authorization.

3. Secure transmission channel

- Following methods are used to secure network communications :
 - a. Virtual Private Network (VPN)
 - b. Hypertext Transfer Protocol Secure (HTTPS)
 - c. Private Network
 - d. Transport Layer Security (TLS) and Secure Socket Layer (SSL)

4. Performance :

Cloud storage performance depends upon speed and latency.

2.6.4 Provisioning Cloud Storage

- Provisioning is the process of setting up IT infrastructure. It can also refer to the steps required to manage access to data and resources and make them available to users and systems.
- Cloud provisioning means allocating a cloud service provider's resource to a customer.
- There are various cloud provisioning delivery models. Each model depends on the types of resources or services an organization purchases, how and when the cloud service provider delivers them and how customers pay for them. The three models : Advanced, dynamic and user self - provisioning.

Cloud provisioning challenges :

1. Complex management and monitoring : Organizations may need several provisioning tools to customize their cloud resources.
- Resource and service dependencies : Cloud applications and workloads often tap into basic infrastructure resources, such as computing, networking and storage. But public cloud service providers offer higher - level ancillary services like serverless functions and Machine Learning (ML) and big data capabilities. Such services may carry dependencies that can lead to unexpected overuse and surprise costs.
 - Policy enforcement : User cloud provisioning helps streamline requests and manage resources but requires strict rules to make sure unnecessary resources are not provided. That is time - consuming since different users require varying levels of access and frequency.

2.6.5 Data Intensive Technologies for Cloud Computing

- Data - intensive systems encompass terabytes to petabytes of data. Such systems require massive storage and intensive computational power in order to execute complex queries and generate timely results.
- Data Intensive Computing is defined as "a class of parallel computing applications which use a data parallel approach to processing large volumes of data".

1. Processing approach :

- Current data - intensive computing platforms use a "divide and conquer" parallel processing approach combining multiple processors and disks in large computing clusters connected using high - speed communications networks.
- It allows the data to be partitioned among the available computing resources and processed independently to achieve performance and scalability based on the amount of data.
- There are several important common characteristics of data intensive computing systems that distinguish them from other forms of computing :
 - a) Data and applications or algorithms are co - located so that data movement is minimized to achieve high performance in data intensive computing
 - b) Programming models that express the high - level operations on data such as data flows are used and the runtime system transparently controls the scheduling, execution, load balancing, communications and movement of computation and data across the distributed computing cluster.
 - c) They provide reliability, availability and fault tolerance.
 - d) They are linearly scalable to handle large volumes of data.

2. System architecture :

MapReduce

- MapReduce is a parallel programming model proposed by Google. It aims at supporting distributed computation on large datasets by using a large number of computers with scalability and fault tolerance guarantees.
- During the map phase, the master node takes the input and divides it into sub - problems, then distributes them to the worker nodes. Each worker node solves a sub-problem and sends the intermediate results ready to be processed by reducer.
- During the reduce phase, intermediate results are processed by reduce function on different worker nodes and the final results are generated.
- Map and reduce are two primitives in functional programming languages, such as Lisp, Haskell, etc. A map function processes a fragment of a key - value pairs list to generate a list of intermediate key - value pairs.

- A reduce function merges all intermediate values associated with a same key and produces a list of key - value pairs as output.
- Properties of Map Reduce :
 - a) Very large scale data.
 - b) MapReduce allows for distributed processing of the map and reduction operations.
 - c) MapReduce provides analytical capabilities for analyzing huge volumes of complex data.
 - d) Number of maps tasks and reduce tasks are configurable.

High Performance Computing Cluster (HPCC) :

- HPCC is also known as DAS (Data Analytics Supercomputer), is an open source, data - intensive computing system platform developed by LexisNexis Risk Solutions. The DAS is a platform designed to refine, link and fuse large amounts of data from disparate sources for complex analysis and queries.
- Architecturally, the DAS is a HPCC based on commodity server hardware, which can be scaled up to thousands of processors to handle any amount of data and runs on the Linux operating system.
- System software and middleware components were developed and layered on to provide the execution environment, distributed file system and SOA interfaces required to support data - intensive computing in an enterprise environment.

2.7 Cloud Storage from LANs to WANs

- The result of the migration from traditional distributed data storage to the cloud computing based on data storage platform.

2.7.1 Cloud Characteristics

- Computer power is elastic, when it can perform parallel operations.
- Data is retained at an unknown host server.
- Data is duplicated often over distant location.

2.7.2 Distributed Data Storage

- A distributed data store is a computer network where information is stored on more than one node, often in a replicated fashion. It is usually specifically used to refer to either a distributed database where users store information on a number of nodes or a computer network in which users store information on a number of peer network nodes.

1. Amazon dynamo :

- Dynamo stores objects associated with a key through a simple interface; it exposes two operations : get() and put().
- The get(key) operation locates the object replicas associated with the key in the storage system and returns a single object or a list of objects with conflicting versions along with a context.
- The put(key, context, object) operation determines where the replicas of the object should be placed based on the associated key and writes the replicas to disk.
- The context encodes system metadata about the object that is opaque to the caller and includes information such as the version of the object. The context information is stored along with the object so that the system can verify the validity of the context object supplied in the put request.
- Dynamo treats both the key and the object supplied by the caller as an opaque array of bytes. It applies a MD5 hash on the key to generate a 128 - bit identifier, which is used to determine the storage nodes that are responsible for serving the key.

2. CouchDB :

- Apache CouchDB is open source database software that has a document - oriented NoSQL database architecture and is implemented in the concurrency - oriented language Erlang; it uses JSON to store data.
- CouchDB is ad-hoc and schema - free with a flat address space. CouchDB comes with a developer - friendly query language and optionally MapReduce for simple, efficient and comprehensive data retrieval.
- Data is stored within JSON documents which can be accessed and its indices queried via HTTP.
- Indexing, transforming and combining of documents are performed through JavaScript. Because it uses all of these web - friendly standards and technologies, CouchDB works very well with web and mobile applications.
- CouchDB is designed to store and report on large amounts of semi - structured, document oriented data.

3. ThruDB

- ThruDB is an open source database built on Apache's Thrift framework and is a set of simple services such as scaling, indexing and storage which is used for building and scaling websites.
- It provides flexible, fast and easy-to-use services that simplify the management of the modern web data layer and provides developers with features and tools most web developers need. These features can be easily configured or turned off.

- Features
 - a) Multi - master replication
 - b) Built for horizontal scalability
 - c) Incremental backups and redo logging
 - d) Multiple storage back - end client libraries for most languages
 - e) Simple and powerful search API
- ThruDB provides web - scale data management by providing these services :
 - a) Thrucene - For Lucene - based indexing
 - b) Throxy - For partitioning and load balancing
 - c) Thrudoc - For document storage
 - d) Thruqueue - For a persistent message queue service
 - e) Thrift - For cross - language services framework

2.8 Multiple Choice Questions

Q.1 NAS is a _____ computer data storage server connected to a computer network providing data access to a heterogeneous group of clients.

- | | |
|--|---|
| ☐ a) file level | ☐ b) directory level |
| ☐ c) root level | ☐ d) all of these |

Q.2 NAS support NFS and _____ file system.

- | | |
|----------------------------------|----------------------------------|
| ☐ a) FAT | ☐ b) NTFS |
| ☐ c) CIFS | ☐ d) KFS |

Q.3 Kosmos File System is an open source project written in _____ by search startup Kosmix.

- | | |
|--|----------------------------------|
| ☐ a) C | ☐ b) Java |
| ☐ c) Machine language | ☐ d) C++ |

Q.4 ThruDB is an open-source database built on Apache's _____ framework.

- | | |
|------------------------------------|--|
| ☐ a) thrift | ☐ b) network |
| ☐ c) file | ☐ d) all of these |

Q.5 Apache CouchDB is open-source database software that has a document-oriented _____ database architecture.

- ☐ a SQL ☐ b NoSQL
☐ c PLSQL ☐ d Oracle

Q.6 Enterprise storage may include _____.

- ☐ a storage area network ☐ b network-attached storage
☐ c direct attached storage ☐ d all of these

Q.7 I/O requests to disk storage on a SAN are called _____

- ☐ a file I/Os ☐ b SAN I/Os
☐ c block I/Os ☐ d disk I/Os

Q.8 Which data storage technology offers the best performance ?

- ☐ a SAN ☐ b NAS
☐ c DAS ☐ d None

Q.9 NFS and CIFS are _____.

- ☐ a file systems ☐ b operating systems
☐ c transport protocols ☐ d file access protocols

Q.10 What's the demerits of DAS ?

- ☐ a Interconnect limited up to 10 km
☐ b Excessive network traffic
☐ c Distance limitations and slow speed
☐ d Distance limitations and inability to share data with other servers

Answer Keys for Multiple Choice Questions :

Q.1	a	Q.2	c	Q.3	d	Q.4	a
Q.5	b	Q.6	d	Q.7	c	Q.8	a
Q.9	d	Q.10	d				



Notes

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